6.1.2 Charge and field

		Learning outcomes	Additional guidance
(a)		*Describe and explain:*	
	(i)	uniform electric field $E = V/d$	*M2.3*
	(ii)	the electric field of a charged object, and the force on a charge in an electric field; inverse square law for point charge	Spherically symmetrical charged conductor is equivalent to a point charge at its centre
	(iii)	electrical potential energy and electric potential due to a point charge; $1/r$ relationship	
	(iv)	evidence for discreteness of charge on electron	Such as the Millikan oil drop experiment HSW2, 7, 11
	(v)	the force on a moving charged particle due to a uniform magnetic field	
	(vi)	similarities and differences between electric and gravitational fields.	HSW2, 8

(b) *Make appropriate use of:*

(i) the terms: charge, electric field, electric potential, equipotential surface, electronvolt

by sketching and interpreting:

(ii) graphs showing electric potential as area under a graph of electric field versus distance, graphs showing changes in electric potential energy as area under a graph of electric force versus distance between two distance values. *M3.8*

(iii) graphs showing force as related to the tangent of a graph of electric potential energy versus distance, graphs showing field strength as related to the tangent of a graph of electric potential versus distance *M3.8*

(iv) diagrams of electric fields and the corresponding equipotential surfaces.

(c) *Make calculations and estimates involving:*

(i) for radial components

$$F_{electric} = \frac{kqQ}{r^2},\ E_{electric} = \frac{F_{electric}}{q} = \frac{kQ}{r^2}\left[k = \frac{1}{4\pi\varepsilon_0}\right]$$

(ii) $E_{electric} = -\frac{\mathrm{d}V_{electric}}{\mathrm{d}r}$,

$E_{electric} = \frac{V}{d}$ (for a uniform field)

(iii) electrical potential energy $= \frac{kQq}{r}$,

$V_{electric} = \frac{kQ}{r}$

(iv) $F = qvB$. *M2.3*